Ella Demarest

Full-Stack Software Engineer | Building AI-powered compliance and audit data systems for regulated industries
San Diego, CA | Open to relocation to SF, NYC, Boston | FT Availability: April 2026 | Open to onsite/hybrid
Ella Demarest Portfolio | LinkedIn | ellajdemarest@gmail.com | (858) 222-7877

EXPERIENCE

Regeneron, Armonk, NY | *Software Engineer* *Intern*: May 2025 – Aug. 2025 | *Contract*: Sept. 2025 – Present

- Designed a relational-vector Postgres database and a TensorFlow-powered semantic search layer using LlamaIndex, enabling similarity analysis across 1,000+ compliance documents to enhance operational efficiency
- Implemented ETL pipelines integrating multiple internal APIs to process 4,000+ compliance-critical documents and 800 DCCs twice daily, feeding into a full-stack React + Flask application that provides real-time reporting for ~40 users to replace manual Excel workflows
- Developed document audit software that identified ~1,200 data quality issues, including 20 critical compliance violations, ensuring high standards for automated data processing
- Automated document role extraction using spaCy NLP, reducing manual review time by 90%

Remix Therapeutics, Watertown, MA | *Software Engineer Co-op* Aug. 2025 – Dec. 2025

- Built a cloud-based large-scale data analysis platform using Python and AWS Nextflow to process high-volume structured datasets and surface correlations via AI
- Developed a Correlation Explorer App using Svelte 5, Vite, and FastAPI for bulk analysis of internal experiments against 4,000+ public datasets

IoT Open Innovation Lab, Northeastern University, Boston, MA | *AI Research Engineer* Oct. 2024 – Apr. 2026

- Engineered a conversational engine for a digital twin simulation app using LLM orchestration, enabling dynamic scenario generation and turn-based user interactions
- Co-founded TwinNU, an AI-powered professional networking platform using digital twin agents to facilitate context-aware relationship maintenance; currently piloting with university research groups

Vue Health, Boston, MA | *Software Engineer Co-op* Jan. 2024 – Oct. 2024

- Built reporting tools used by client-facing teams to present structured data to external stakeholders
- Engineered client-facing workflow systems integrated with enterprise CRM to support operations and reporting

PROJECTS

LinesLight: MBTA V3 API (React, TypeScript, & Stripe) Boston, MA | Dec. 2025

- Engineered a responsive web app (https://lineslight.live/) utilizing the MBTA V3 API to enable A to B trip planning and live vehicle tracking, providing Boston commuters with real-time arrival predictions and optimized navigation of departure data

RBLXStats Analytics Platform (React, TypeScript, PostgreSQL, & Docker) Boston, MA | Oct. 2025

- Built a production-grade analytics platform (https://rblxstats.live) with hourly sampling workers, shared API polling, and 90-day retention backed by PostgreSQL 15 with monthly partitioning, precomputed aggregates, and fast TypeScript/Express API endpoints powering real-time React + Chart.js dashboards
- Integrated an AI insights agent that analyzes 24-hour performance windows to surface trends and retention diagnostics

TECHNICAL SKILLS

Languages: Python | TypeScript | JavaScript | SQL | HTML | CSS
Frontend: React | Tailwind | Vite | Chart.js
Backend: Node.js | FastAPI | Express | Flask | Stripe API | REST APIs
Databases & Data Warehousing: MongoDB | PostgreSQL | MySQL | Redshift
Cloud, Tools & Platforms: AWS | GCP | Azure | Firebase | Docker & Docker Compose | Git/GitHub | Postman
Methodologies & Practices: Agile | CI/CD pipelines
AI & Machine Learning: LLM Orchestration | NLP (spaCy) | vector databases (pgvector) | PyTorch | TensorFlow

EDUCATION

B.S. Computer Science, Northeastern University, Boston, MA Expected April 2026
Computer Science Concentration: Human-Centered Computing GPA: 3.84/4.0